Optoelectronic RP-Style Decision Circuit

Mathematically corrected edition • Revised equations, probability bounds, photon statistics, and feedback interpretation.

1. Scope and mathematical status

This document describes an intended randomized decision architecture. The block labels alone do not prove that the hardware implements an RP algorithm, produces nonclassical light, or realizes a closed timelike curve. Those claims require separate algorithmic proofs, electrical specifications, and physical evidence.

2. Definition of RP

Let L be the language being decided, x an input, and r the random bits of a trial. A randomized algorithm M has one-sided error when:

$$x \notin L \;\Rightarrow\; \Pr_r[M(x,r)=1]=0$$

$$x \in L \;\Rightarrow\; \Pr_r[M(x,r)=1] \geq \frac{1}{2}$$

For membership in RP, the worst-case running time must be polynomial in the input length for every random choice. A fixed 10 Hz sampling clock does not establish polynomial-time scaling.

3. Intended processing chain

Input x → randomized witness test A and deterministic precondition B → explicit AND logic → defined signal conditioning → enabled sampling → decision memory. An optical detector may supply randomness, but its presence does not prove the witness-test guarantees.

The original drawing should be treated as a conceptual block diagram, not a construction-ready schematic. This edition replaces it with explicit logical and statistical specifications rather than reproducing misleading electrical connections.

Source basis: user-supplied seven-page analysis; the corrections discussed in the accompanying conversation. External references are listed on page 7.

Driver Logic and Sampling

4. Driver definitions and acceptance

$$A = w(x, r), \qquad B = c(x), \qquad A, B \in \{0, 1\}$$

$$D = (A \vee B) \wedge (A \wedge B) = A \wedge B$$

This identity assumes the OR and AND results are combined by a further logical AND. It does not describe directly tying push-pull gate outputs together.

$$\Pr_r[D = 1 \mid x] = c(x)\Pr_r[w(x, r) = 1]$$

Driver A must never accept a NO instance and must accept every YES instance with probability at least one half. Driver B must equal 1 on every YES instance; otherwise it can permanently veto a valid input.

Input / event	A	B	D
NO instance	0	0 or 1	0
YES; witness found	1	1 required	1
YES; witness missed	0	1 required	0

5. Time code must enable, not override

$$D_j = A_j \wedge B_j \wedge Y_j$$

$$(Y_1, Y_2, Y_3, Y_4) = (1, 0, 1, 1)$$

A time-code bit may mask a trial. It must not independently force D HIGH: OR-ing Y with the acceptance condition would permit false acceptance. Count only enabled, completed trials in error-reduction bounds.

6. Flip-flop rule

$$Q(t_j^+) = D(t_j^-), \qquad \overline{Q}(t_j^+) = 1 - Q(t_j^+)$$

This is the ideal sampling rule at rising clock edge j, subject to setup/hold requirements and inactive asynchronous controls. The complementary output is not an independent observation channel.

Independent Trials and Error Bounds

7. Accumulate acceptance across trials

$$S_0 = 0, \qquad S_j = S_{j-1} \vee D_j$$

$$S_k = \bigvee_{j=1}^{k} D_j$$

Repetition reduces rejection error only when the final decision accepts if any trial accepts. A flip-flop that merely stores the latest trial does not implement this accumulation.

For independent trials on a YES instance, let the acceptance probability on trial j be p_j, with each p_j at least one half.

$$\Pr[S_k = 0 \mid x \in L] = \prod_{j=1}^{k} (1 - p_j) \le 2^{-k}$$

$$p_j = p \;\Rightarrow\; \Pr[S_k = 0 \mid x \in L] = (1-p)^k$$

$$\Pr[S_k = 1 \mid x \notin L] = 0$$

The worst-case bound is at most 2 to the power –k; equality requires the limiting case p_j = 1/2 on every trial. Fresh independent randomness is an assumption, not a consequence of waiting for another clock edge.

8. Nominal 10 Hz timing

$$f = 10\,\text{Hz}, \qquad T = 1/f = 0.1\,\text{s}, \qquad t_k = kT$$

Trials k	Rejection upper bound	Nominal elapsed time
1	0.5	0.1 s
2	0.25	0.2 s
5	0.03125	0.5 s
10	0.0009765625	1.0 s
20	0.00000095367431640625	2.0 s

Elapsed times assume one completed, enabled trial per period and no preparation or processing overhead. Disabled time-code windows increase elapsed time without adding trials.

Photon Statistics and Thinning

9. Definitions with distinct notation

Use Q for the flip-flop output and Q_M for the Mandel parameter. All photon-count moments below refer to the same specified mode or counting bin, with positive mean count.

$$\mu = \langle n \rangle > 0, \qquad F = \frac{\mathrm{Var}(n)}{\mu}, \qquad Q_{\mathrm{M}} = F - 1$$

$$g^{(2)}(0) = \frac{\langle n(n-1) \rangle}{\mu^2}$$

$$F = 1 + \mu[g^{(2)}(0) - 1], \qquad Q_{\mathrm{M}} = \mu[g^{(2)}(0) - 1]$$

These identities follow from expanding the factorial moment. A finite-window count statistic must not be silently equated to a differently defined time-resolved correlation measurement.

Ideal count model	F	g²(0)
Poisson distribution	1	1
Single-mode thermal distribution	1 + mean count	2
Exactly one photon in the specified mode	0	0

10. Independent carrier-to-photon thinning

$$N_{\mathrm{ph}} \mid N_e \sim \mathrm{Binomial}(N_e, \eta)$$

Assume each carrier independently yields at most one detected photon with the same probability eta, without added counts or correlations.

$$\mathbb{E}[N_{\mathrm{ph}}] = \eta \mathbb{E}[N_e]$$

$$\mathrm{Var}(N_{\mathrm{ph}}) = \eta(1-\eta)\mathbb{E}[N_e] + \eta^2 \mathrm{Var}(N_e)$$

$$F_{\mathrm{ph}} = 1 - \eta + \eta F_e = 1 - \eta(1 - F_e)$$

Thus F_e = 0 and eta = 0.8 give F_ph = 0.2. F_e = 1 implies F_ph = 1, but a unit Fano factor alone does not prove the full distribution is Poissonian. An ordinary NPN threshold stage does not establish deterministic single-carrier injection.

Detection, Transitions, and Measurement

11. Detection opportunities and dark counts

If every trial supplies a new independent detection opportunity with success probability eta, the probability of missing all opportunities is:

$$P_{\text{miss},\,k} = (1-\eta)^k$$

This assumes new opportunities; it does not describe repeated detection attempts on the same absorbed photon. For independent dark-click probability d per window:

$$P_{\text{false accept},\,k} = 1-(1-d)^k$$

$$1-(1-d)^k \approx kd \qquad (kd \ll 1)$$

Expected dark-count totals can grow linearly, but the probability of at least one dark click is not exactly linear. Nonzero dark clicks break a literal zero-false-accept detector analogy.

12. Correct transition directions

$$E_1 \longrightarrow E_2 : \quad \text{excitation requires energy } E_2 - E_1$$

$$E_2 \longrightarrow E_1 : \quad h\nu = E_2 - E_1$$

Upward excitation requires supplied energy. Downward radiative decay emits a photon. The original switching table incorrectly assigned emission to the upward transition.

13. Ideal coherent excitation

$$P_e = \sin^2(\theta/2), \qquad \theta = \pi \;\Rightarrow\; P_e = 1$$

This expression assumes the ideal resonant two-level pulse model, starting in the ground state and neglecting decay and dephasing during the pulse. Labeling a classical driver deterministic does not establish these conditions.

14. Number measurement versus clicks

$$P(n) = |\langle n|\psi\rangle|^2$$

This is the ideal number-measurement probability for a pure single-mode state. A binary click/no-click detector is not generally photon-number resolving; its click histogram is not automatically the full photon-number distribution. Detection efficiency, dark counts, and detector response must enter the measurement model.

Feedback and CTC Consistency

15. Ordinary feedback is not backward time travel

Returning Q to a driver can condition future inputs. That causal feedback is not, by itself, a closed timelike curve and does not enforce Deutsch consistency.

16. Deutsch fixed-point equation

$$\rho_{\mathrm{CTC}} = \mathrm{Tr}_{\mathrm{other}}[U(\rho_{\mathrm{other}} \otimes \rho_{\mathrm{CTC}})U^{\dagger}]$$

This equation defines consistency in a hypothetical CTC model. It is not derived merely by adding an electronic feedback wire.

17. Classical NOT versus quantum NOT

$$p = 1 - p \quad \Rightarrow \quad p = \tfrac{1}{2}$$

A classical NOT map has a consistent probability distribution assigning equal probability to 0 and 1. For a quantum NOT channel, however, the full state must satisfy:

$$\rho = X\rho X, \qquad \rho = \tfrac{1}{2}(I + r_x X + r_y Y + r_z Z)$$

$$r_y = r_z = 0, \qquad -1 \le r_x \le 1$$

There is a family of quantum fixed points. I/2 is the maximum-entropy member, not the unique fixed point. Its equal probabilities do not derive the RP success threshold.

18. Complexity theorem and its limitation

$$\mathrm{P}_{\mathrm{CTC}} = \mathrm{BQP}_{\mathrm{CTC}} = \mathrm{PSPACE}$$

The Aaronson–Watrous theorem concerns polynomial-time computation equipped with Deutsch-style causal-consistency resources. It does not show that ordinary feedback circuits solve PSPACE problems, and it does not establish a strict separation between RP and PSPACE.

Implementation Boundaries and References

19. Electrical statements that need specification

An RC time constant is tau = RC only for a defined equivalent resistance and capacitance. A 10 microfarad label alone does not specify a cutoff frequency or settling time. Charging, loading, threshold crossings, reset/discharge paths, and sampling margins must be analyzed for the actual circuit.

$$\tau = RC, \qquad f_c = \frac{1}{2\pi RC}$$

The cutoff expression applies to an ideal first-order RC low-pass network. A transistor base-emitter voltage near 0.7 V is an approximation, not an exact universal threshold. An emitter-follower output is not automatically driven to the collector supply voltage.

Do not directly join ordinary push-pull OR and AND outputs to implement a Boolean operation. Specify a real combining gate or a compatible interface. Specify inactive asynchronous flip-flop inputs, supply decoupling, input voltage limits, and setup/hold margins using the selected component datasheets.

20. Revised conclusion

The intended abstract decision rule is A AND B, optionally masked by Y. It has RP-style one-sided error only when the witness test and deterministic precondition meet their stated guarantees. Independent trials combined by an acceptance OR reduce rejection probability to at most 2 to the power –k. The thinning formula is conditional on independent Bernoulli conversion. Neither nonclassical emission nor a CTC computational resource follows from the pictured block labels.

21. References and source notes

User-supplied source: *Optoelectronic RP Decision Circuit — Complete Analysis*, seven supplied pages. This edition revises its mathematical claims and replaces its conceptual schematic with explicit specifications.

S. Vadhan, *The Power of Randomness*, Definition 2.9 (RP): people.seas.harvard.edu/~salil/pseudorandomness/power.pdf.

S. Aaronson and J. Watrous, “Closed timelike curves make quantum and classical computing equivalent,” *Proceedings of the Royal Society A* 465 (2009), 631–647. Author manuscript: cs.uwaterloo.ca/~watrous/Papers/ClosedTimelikeCurves.pdf.

D. Deutsch, “Quantum mechanics near closed timelike lines,” *Physical Review D* 44 (1991), 3197–3217.

Photon-count identities and the thinning relation are derived from the moment definitions and the conditional binomial model stated on page 4. Detector-window bounds follow from independent Bernoulli trials.